INCH-POUND

MS25031N
26 August 2020
SUPERSEDING
MS25031M
27 November 2003

DETAIL SPECIFICATION SHEET

RELAY, ELECTROMAGNETIC,100 AMPERES, 2 PDT, N.O. TYPE II, NON-HERMETICALLY SEALED, MECHANICALLY INTERLOCKED

Inactive for new design effective 28 September 1990.
No superseding specification.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

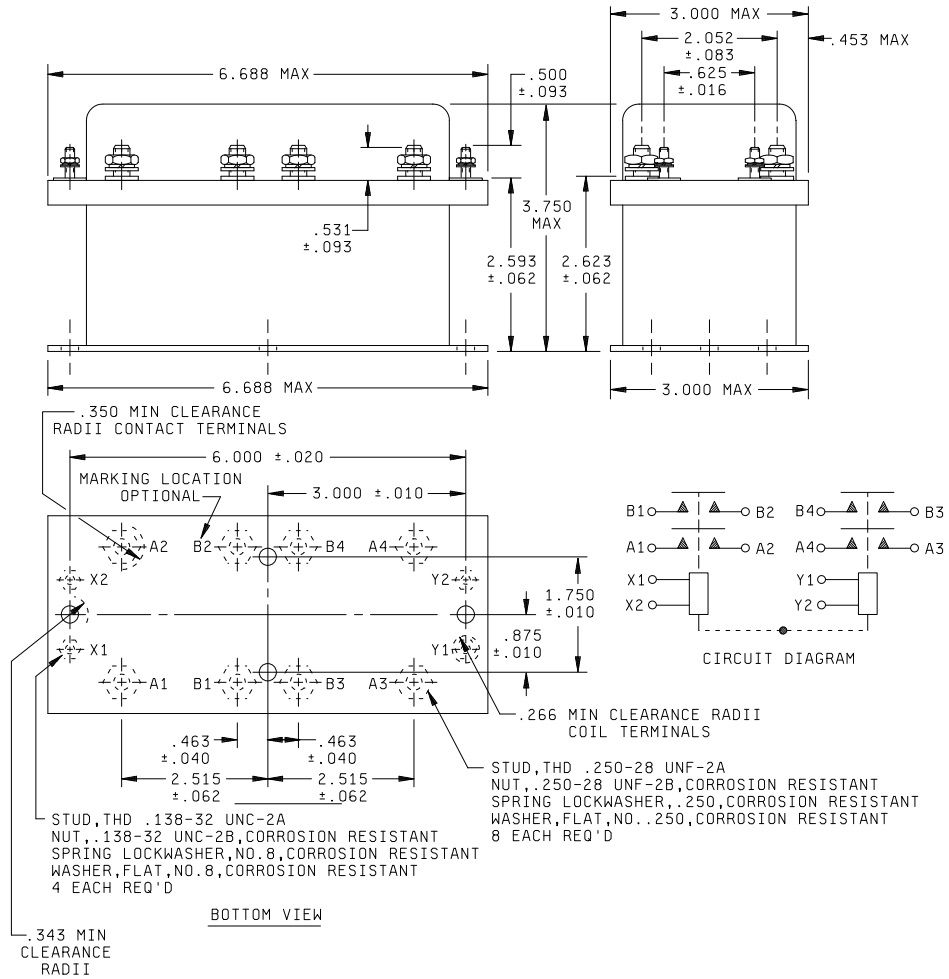


FIGURE 1. Dimensions and configurations.

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Inches	mm	Inches	mm
.010	0.25	.500	12.70
.020	0.51	.531	13.49
.040	1.02	.810	20.57
.062	1.57	.875	22.23
.083	2.11	1.750	44.45
.093	2.36	2.052	52.12
.138	3.51	2.215	63.88
.201	5.11	2.593	65.86
.250	6.35	2.623	66.62
.266	6.76	3.000	76.20
.343	8.71	3.750	95.25
.350	8.89	6.000	152.40
.463	11.76	6.688	169.88

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.032 (0.81 mm).
4. Additional flat washer may be used for terminal seat.
5. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
6. Referenced Government documents of the issue listed in Assist Online (https://assist.dla.mil/) or Assist Quick Search (https://quicksearch.dla.mil/) specified in the solicitation form a part of this specification to the extent specified herein.
7. Terminal numbers shall not appear on relay header. There shall be affixed to the relay, a suitable legible circuit diagram that identifies each terminal location.
8. Mechanical linkage shall be provided to prevent both relays being closed at the same time.

FIGURE 1. Dimensions and configurations - Continued.

TABLE I. Dash numbers and characteristics.

Dash number MS25031-	Type	Coil	Terminal type	Mounting or mating socket	Max weight in pounds
D1B	II	dc	Stud	Plate	3.5

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TABLE II. Operating characteristics.

PIN MS25030-	Coil data												Time - milliseconds max				
	Coil	Rated			Max		1/ Max pick-up voltage			Hold voltage 2/	Drop out voltage 2/	Operate 3/	Release 4/	Bounce			
		Volts 1/	Freq Hz	Ω Res	Volts	Amp	Normal 2/	High temp test	Cont current test					Main		Aux	
														NO	NC	NO	NC
D1B	X1, X2 Y1, Y2	28	dc	47.5	29	.70	18	21	22.5	7.0	1.5	35	15	3.5	N/A	N/A	N/A

1/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/			
		Main		Aux		Main		Aux		Main		Aux	
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz
Resistive	50	100				75							
Inductive	10	80											
Inductive													
Motor	50	80				65							
Lamp													
Transfer load 2/													
Mechanical life reduced current	100	25				25							
Mixed loads	50	Applicable in accordance with MIL-PRF-6106											

1/ Absence of value indicates relay is not rated for 3-phase application.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.

Environmental characteristics.

Temperature range	-55°C to +71°C
Max altitude rating	50,000 ft
Shock G-level	25 g's
Duration	6-9 ms
Max duration contact opening	2 ms
Vibration - sinusoidal (see chart below)	
G-level	g
Frequency range	Hz
Vibration - random	
Applicable spec	N/A
Power spectral density RMS G min	N/A
Frequency range Curve	N/A
High shock	N/A
Acceleration	10 g's

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Electrical characteristics.

Minimum insulation resistance, initial	100 megohms.
After life or environmental tests	50 megohms.
Dielectric strength (sea level).	2-5 seconds.

	Initial		After life tests	
	28 V dc	115 V ac	28 v dc	115 V ac
Coil to case	1,250 V rms	N/A	1,000	N/A
Aux contacts	1,250 V rms	N/A	1,000	N/A
All other points	1,250 V rms	1,500	1,000	1,125

Dielectric strength (altitude) 1 minute.

	Initial	After life tests
Coil to case	500	N/A
Aux contacts	500	N/A
All other points	500	500

Max contact drop initial	0.150 volt.
After life test	0.175 volt.
Overload current (NO)	800 amperes.
Rupture current	1,000 amperes.
Duty rating	Continuous.
RFI specification	
(Applicable to coil circuits of ac operated relays).	

Conformance inspection.

Performance of groups B and C tests are not applicable.

Group A acceptance reports shall be submitted to the preparing activity on a yearly basis in order to retain qualification for this military standard sheet.

Qualification by similarity: See MIL-PRF-6106.

Dash number MS25031-	Vibration level				
	5-10 Hz	10-55 Hz	55-250 Hz	250-500 Hz	500-1500 Hz
D1B	.08 DA	.06 DA	2 g's	2 g's	

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. This document references MIL-PRF-6106.

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2020-047)

Review activities:
Navy - EC

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